Airflow Masterclass

What is Airflow?

Airflow is an open source framework which can be used as an ORCHESTRATOR.

What Airflow Is Not?

Airflow is not a DATA PROCESSING framework.

Airflow is not a REALTIME PROCESSING framework.

Airflow is not an ETL framework.

Why not Script?

Core Components of Airflow

Metadata DB

Airflow needs to store all the metadata about the DAGs. For instance, DAG runs, schedule, status, task instance etc.

DAG File Processor

DAG File processor will go to your dags folder and will parse it and store the serialized DAG into the DB

{"xyz": "abc"}

API Server

Scheduler

Scheduler is responsible for WHAT & WHEN tasks need to be executed.

Executor

Executor will decide the HOW & WHERE tasks need to be run.

Workers

Workers are the real layer for execution.

Queue

Triggerer

Airflow Architecture (Imp)

Core Concepts of Airflow

DAG

Directed Acyclic Graph

Task Instance

It's just a single unit of work.

Operator

It's the pre-built template for our tasks.
It basically tells Airflow what type of task we are running.

How DAGs are synced?

XCOMs In Airflow

Parallel Tasks

Scheduling In Airflow

start_date (Required)

end_date (optional)

catchup (optional)

It is used to tell Airflow to run the DAG for the missed intervals.

schedule (Required)

Presets: @daily, @hourly, @weekly

Cron: 0 16 * * MON-FRI

Delta: 1,4,7,10,13,16,19,22,25,28,31

Feb: 1,4,7,10,13,16,19,22,25,28,31
3,6,9,12,15,18,21,24,27

Incremental Load & Templating

Special Schedules - Events

Assets

DAGs Orchestration